Pranav Minasandra, Ph.D.

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Interests

I am a behavioural theorist and ethologist. My work explores the advantages of acknowledging the inherent complexity of behaviour in models, which sometimes leads to surprising insights. Broadly, I model behavioural transition dynamics across species. I also perform simulations to study how behaviour and cognitive abilities impact movement. Part of my research also concerns the use of more advanced programming methods in behavioural ecology.

Experience

Postdoc | **Max Planck Institute of Animal Behaviour, Konstanz, Germany** — 2024 - now
IMPRS-QBEE Postdoc Awardee.

Ph.D. | **Max Planck Institute of Animal Behaviour, Konstanz, Germany** — 2020 - 2024
Magna cum laude
Thesis: *"Exploring sequences of behaviour across scales and species."*
Also affiliated to: International Max Planck Research School of Quantitative Behaviour, Ecology, and Evolution; and the University of Konstanz.

M.Sc. | **Indian Institute of Science (IISc), Bengaluru, India** — 2019 - 2020
Master of Science in Biology.
Research at the Centre for Ecological Sciences, IISc.

B.Sc. (Research) | **Indian Institute of Science (IISc), Bengaluru, India** — 2015 - 2019
Major in Biology.
Research at the Centre for Ecological Sciences, IISc.

Publications

Peer-reviewed publications

1. Grout, E. M., Hass, C. C., Minasandra, P., Thomas, M., Ashbury, A. M., Ortega, J., Crofoot, M. C., Hirsch, B. T. & Strandburg-Peshkin, A. The vocal repertoire of white-nosed coatis: structural features, temporal dynamics, and social variations. *Bioacoustics*. 1–23. doi:10.1080/09524622.2026.2714161 (2026).
2. Gall, G. E., Demartsev, V., Minasandra, P., Baldoni, C., Cain, K. E. & Quinn, J. S. Examining Combinatoriality within the Pūkeko Vocal Repertoire. *Animal Behaviour*. 123431. doi:10.1016/j.anbehav.2025.123431 (2026).
3. Minasandra, P., Grout, E. M., Brock, K., Crofoot, M. C., Demartsev, V., Gersick, A. S., Hirsch, B. T., Holekamp, K. E., Johnson-Ulrich, L., Nayak, A., *et al.* Behavioral sequences across multiple animal species in the wild share common structural features. *PNAS* **122.** e2503962122. doi:10.1073/pnas.2503962122 (2025). [Best Paper Award | Press coverage].
4. DeNicola, V., Mezzini, S., Bursac, P., Minasandra, P. & Cagnacci, F. Effects of vasectomy on breeding-related movement and activity in free-ranging white-tailed deer. *Movement Ecology*. doi:10.1186/s40462-025-00554-5 (2025).
5. Grout, E. M., Ortega, J., Minasandra, P., Quin, M. J., Crofoot, M. C., Strandburg-Peshkin, A. & Hirsch, B. T. Whole Group Tracking Reveals That Relatedness Drives Consistent Subgrouping Patterns in White-Nosed Coatis. *Animal Behaviour* **216.** 175–193. ISSN: 0003-3472. doi:10.1016/j.anbehav.2024.08.010 (2024).
6. Minasandra, P., Jensen, F. H., Gersick, A. S., Holekamp, K. E., Strauss, E. D. & Strandburg-Peshkin, A. Accelerometer-based predictions of behaviour elucidate factors affecting the daily activity patterns of spotted hyenas. *Royal Society Open Science* **10.** 230750. doi:10.1098/rsos.230750 (2023).

7. Kunjar, S., Strandburg-Peshkin, A., Giese, H., Minasandra, P., Sarkar, S., Jolly, M. K. & Gradwohl, N. Link updating strategies influence consensus decisions as a function of the direction of communication. *Royal Society Open Science* **10.** 230215. doi:10.1098/rsos.230215.
8. Minasandra, P. & Isvaran, K. Truncated power-law distribution of group sizes in antelope. *Behaviour* **157.** 541–558. doi:10.1163/1568539X-bja10012 (2020).

Pre-prints and work in-prep

9. Minasandra, P. Anticipating others' future behaviours alters collective movement and enhances survival. *biorxiv* **122.** 2025–08. doi:10.1101/2025.08.14.670290 (2025). [In review].
10. Minasandra*, P., Sridhar*, V. H., Roche, D. G. & Planas-Sitjà, I. TracktorLive: An Integrated Real-Time Object Tracking and Response System. *bioRxiv*. 2026.03.12.711471. doi:10.64898/2026.03.12.711471 (2026). [In review, GitHub].
11. Gall, G. E. C., Minasandra, P., Cain, K. E. & Quinn, J. S. Ontogeny of vocal sound sequences in pūkeko (*Porphyrio melanotus*) (2026). [In review].
12. Minasandra, P., Demartsev, V., Johnson-Ulrich, L., Manser, M. & Strandburg-Peshkin, A. Spatiotemporal interplay of vigilance, movement, and collective behaviour in groups of meerkats. [Ch. 3 of Dissertation, available online].
13. Goldsborough, Z., Crofoot, M., Minasandra, P., León, J., Castillo-Caballero, P. L., del Rosario-Vargas, E. & Barrett, B. J. Habitual tool use on monopolizable resources affects group cohesion. doi:10.32942/X25H1Q (2025). [In review].

(* → co-first author)

Mentorship

Oakleigh Wilson (Ph.D. student) — 2026 - now
Role: External Advisor
Doctoral thesis chapter: Behaviour inference from accelerometer data through unsupervised machine learning.

Juee Dhar (Ph.D. student) — 2025 - now
Role: Mentor and Thesis Advisory Committee member.
Doctoral thesis: Sleep social dynamics in baboon groups.

Amlan Nayak (Master's student) — 2022 - 2023
Role: Mentor and primary co-advisor.
Master's thesis: Spatial variation of vigilance in meerkats.

Ananya Passi (Bachelor's student) — 2019
Role: Co-advisor
Summer research: Mathematical modelling of habitat use and population dynamics.

Awards and fellowships

ISBE Travel Award — 2026
Registration fees and travel costs to ISBE conference 2026. [≈1,600 €]

Best Paper Award—University of Konstanz — 2025
Chosen from among several nominations. [200 €]

Nominated—Schmidt Science Fellows programme — 2024
One of ten doctoral graduates nominated by the Germany-wide Max Planck Society.

IMPRS Postdoctoral Award — 2024 - 2025
6 months of pay at the scale of a postdoctoral researcher. [32,350 €]

ISBE Travel Award — 2024
2500 AUD [≈1,500 €].

DAAD—Graduate Student Scholarship Programme 2020 - 2024
Nominated after evaluation of my proposal. [≈60,000 €]

Kishore Vaigyanik Protsahan Yojana 2015 - 2020
Govt. of India scholarship for exceptional undergraduates in science. *All India Rank : 135*

Talks and posters

[Invited] International Conference of Neuroethology 2026
Vancouver, Canada
"The Behavioural Observatory: Cross-species comparison of continuous decision-making"
(in the invited symposium *Brains in the Wild* by Jan Benda.)

International Society of Behavioural Ecology Meeting 2026
Turin, Italy
"The Behavioural Observatory: Cross-species comparison of continuous decision-making"

[Invited] Talk at Laboratory for the Conservation of Endangered Species 2025
Centre for Cellular and Molecular Biology, Hyderabad
"Three short stories in animal behaviour"

[Invited] CES Seminar Series 2024
Indian Institute of Science, Bangalore
"Exploring sequences of behaviour in the wild across scales and species"

International Society of Behavioural Ecology Meeting 2024
Melbourne, Australia
"Sequences of behaviour show a common statistical structure across species"

Gesselschaft für Primatologie Annual Conference 2024
Konstanz, Germany
"How animals behave: Sequences of behavior show a common statistical structure across species"

Descriptive & Normative Models of Collective Behaviour 2022
School of Mathematics, University of Leeds
"Looking into hyena daily activity patterns using accelerometers"

Teaching

Introduction to Git ongoing
Instructor and organiser: Hands on in-house workshop. [link to class notes]

Introduction to remote code servers like GitHub ongoing
Instructor and organiser: Hands on in-house workshop. [link to class notes]

LaTeX for academic documents ongoing
Instructor and organiser: Hands on in-house workshop. [link to class notes]

Quantitative Ecology: Research Design and Inference 2019
Teaching assistant: Taught coding in R, conducted several classes, graded assignments, managed course website.

Technical skills

Programming languages and related
- Python, Julia, R, Bash, LaTeX, and C.
- HTML, CSS, and JavaScript.
- Version control with Git; GitHub (including actions, workflows, and organisation management).
- Tools for Linux server setup and management.
- Compiled and hosted a Linux from Scratch OS.

Artificial Intelligence (AI)
- Agentic code generation and validation.

- Hosting local Large Language Models (LLMs); data inference with LLMs (with `ollama`); vision-capable LLMs and video processing;
- Design of basic agentic loops.

Other specialised computational skills
Machine learning; Parallel processing and Inter-Process Communication; Agent-based models; Analysis of complex multivariate time-series data.

Mathematical modelling
Dynamical (PDE) models incorporating spatial and stochastic variables; Probability models; Numerical simulations of all the above.

Other academic services

Reviewing
Animal Biotelemetry; Ecology and Evolution; Methods in Ecology and Evolution; PLOS One; Integrative and Comparative Biology; Animal Behaviour.

Representation 2024 - 2025
Postdoc representative at the Max Planck Institute of Animal Behaviour.